

65. Conceptual

66. Conceptual

67. Sb = Bi

68. $X = 2(CH_3CHO, HCOOH)$

$Y = 4(C_2H_5OH, CH_3CHO, CH_3COCH_3, CH_3CHOHCH_3)$

= ( )

69.

( ) → ( ) + ( )

– –

(–)

630 + 2p = 1260

2p = 630

P = 315

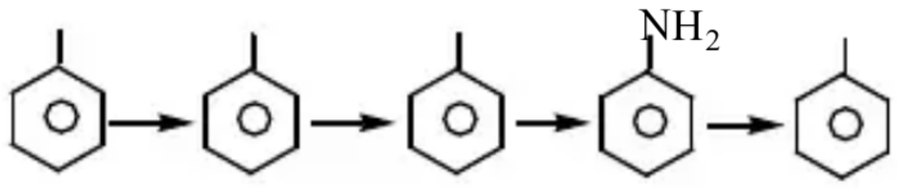
/ = minutes

$K = \frac{0.693}{50} = 0.01386\,min^{-1}$

70. RnF_2- detected by spectroscopic technique

XeF_2-linear (planar) molecule

71.



72. (I) MOT (II) Ionic, covalent, H-bonds

73. None of them are feasible.

Acidic strength of

2 3 > 6 5 > 2 > 2 5 > 2 2

74. $\Delta^0G = -T\Delta^0S_{total}$

Δ =–

$-2.303\ RT\ \log\ K_{eq} = -T\Delta^0S_{tot}$

$\Delta^0S_{tot} = 2.303\ R \log k_{eq}$

$= 2.303(R)\log 10^2$

$= 4.606R$

75.

= +

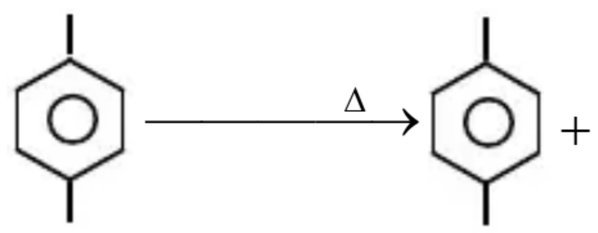
∴ (–) = – ( ) = –

+ = – + ( ) –

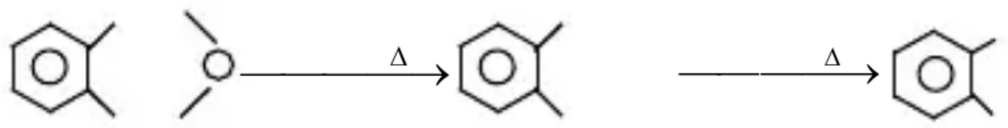
= –

$= -0.2364 - (-0.815) = 0.579$

76.

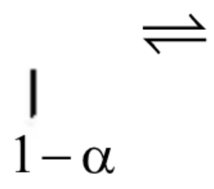


X = 2



y =2

77.



1 – α 2α

$= \frac{\left(\frac{\alpha}{+\alpha}\right)}{\left(\frac{-\alpha}{+\alpha}\right)} = \frac{\alpha}{(+\alpha)} \times \frac{(+\alpha)}{(-\alpha)} = \frac{\alpha}{-\alpha}$

$= \frac{(\quad)}{-}$

$= \frac{(\quad)}{\quad}$

=

78. $X = 3^0$ butyl alcohol

79. Conceptual

80.

+ – → + – Δ = –

+ + – → Δ – –

→ + + – Δ = +

81. Conceptual

82. $2H_2O_2 \xrightarrow[OH^-]{I^-} 2H_2O + O_2$

$\text{Rate} = K[H_2O_2][I^-]$

83.

$LiAlH_4$ **reduces** $-COOH \rightarrow -CH_2OH$ **without reducing C = C**

PCC **oxidises** $-CH_2OH \rightarrow -CHO$ **without oxidising C = C**

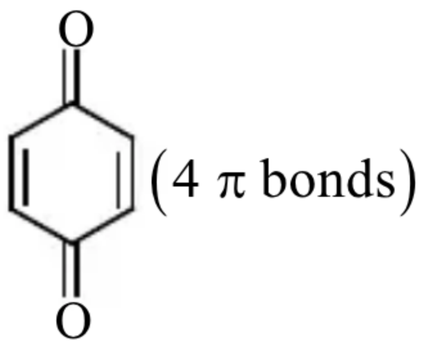
84. $Fe_2(C_2O_4)_3 \longrightarrow Fe^{3+} + CO_2$

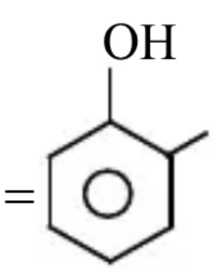
$EW = \frac{M}{6}$

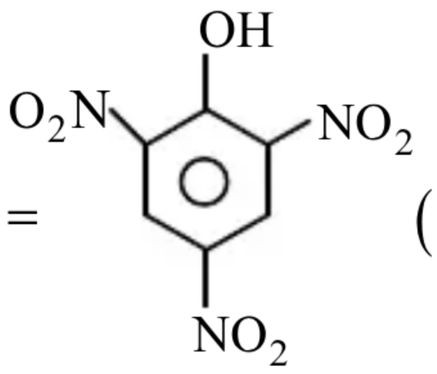
85. $C_4H_{10} + \frac{13}{2}O_2 \longrightarrow CO_2 + 5H_2O$

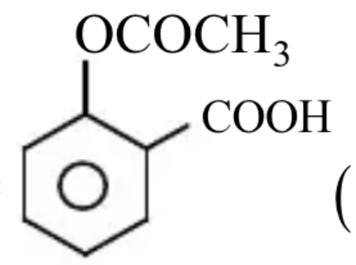
$n = 26$

86.

P =  (4 π bonds)

Q =  (4 π bonds)

R =  (6 π bonds)

S =  (5 π bonds)

87. $1.06 \times 10^{-8} = \frac{6.625 \times 10^{-34}}{4 \times 3.12 \times m}$

$m = \frac{0.53 \times 10^{-34}}{1.06 \times 10^{-8}} = 0.5 \times 10^{26} = 5 \times 10^{-27}$

88. Glucose → Saccharic acid → Hexane

89. $x = 4$

$4 = 40 \times 10^3 \left(\frac{n_{Ar}}{\frac{500}{18}} \right); n_{Ar} = \frac{500 \times 4}{40 \times 10^3 \times 18} = \frac{1}{360}$

90. Conceptual

BIOLOGY

91. Refer XI NCERT Page No – 104

94. Refer XI NCERT Page No – 317

96. Refer XII NCERT Page No – 32

At least 60% of sperms must have normal shape and size and at least 40% of them must show vigorous motility.

98. Refer XII NCERT Page No – 45

99. Refer XI NCERT Page No – 306

101. Refer XI NCERT Page No – 56

Notochord is persistent through out life in cephalochordates and Chondrichthyes.

102. Refer XI NCERT Page No – 312

104. Refer XI NCERT Page No – 47

106. Refer XII NCERT Page No – 133

107. Refer XI NCERT Page No – 293

Filtration membrane is made of 3 layers. But basement membrane is non-cellular.

110. Refer XI NCERT Page No – 270

113. Refer XII NCERT Page No – 111

Coacervates does not have definite lipid membrane and reproduction also absent.

114. Refer XI NCERT Page No – 285

Heart rate is less in athletes when compared to non-athletes under resting conditions.

116. Refer XI NCERT Page No – 280, 281

117. Refer XI NCERT Page No – 119

No.of cranial nerves in Frog is – 10 pairs (20)

120. Refer XII NCERT Page No – 76

122. Refer XII NCERT Page No – 201

Interaction between red billed oxpecker and black rhinoceros is example of proto co-operation.

124. Refer XII NCERT Page No – 121

125. Refer XI NCERT Page No – 114

126. Refer XI NCERT Page No – 287, 288

Asprin inhibits vaso – constriction and platelet aggregation.

129. Refer XI NCERT Page No – 268, 269

131. Refer XII NCERT Page No – 120, 121

Total pop – 400

recessive individuals – 64.

$$q^2 = \frac{64}{400} = 0.16 \quad q = 0.4$$

$$2pq = 0.48$$

Total percentage of heterozygous individuals – 48%.

133. Refer XI NCERT Page No – 53, 54

134. Refer XI NCERT Page No – 334, 335

136. Refer XI NCERT Page No – 49

139. Refer XII NCERT Page No – 38, 39

141. Refer XI NCERT Page No – 73

The given pedigree chart is example of autosomal recessive.

143. Refer XII NCERT Page No – 37

Morula is same size of Zygote (or) fertilized egg.

144. Refer XI NCERT Page No – 297

146. Refer XII NCERT Page No – 134, 135

148. Refer XII NCERT Page No – 106

151. Refer XII NCERT Page No – 119

152. Refer XII NCERT Page No – 190, 191

155. Refer XII NCERT Page No – 138

156. Refer XII NCERT Page No – 181, 182

158. Refer XII NCERT Page No – 224, 225

160. Refer XII NCERT Page No – 141, 142

161. Refer XII NCERT Page No – 224

162. Refer XII NCERT Page No – 134, 135

Natural killer cells not help in phagocytosis they are non-phagocytic cells.

166. Refer XII NCERT Page No – 209

If R is – 150 k.J, then R = 25 %

$$\frac{x}{4} = 150 \; X = 150 \times 4 = 600kJ$$

$$X = GPP\left(600kJ\right)$$

$$GPP - R = 450kJ$$

So, N.S.P in secondary carnivores is – 0.45 kJ

167. Refer XI NCERT Page No – 333

168. Refer XII NCERT Page No – 46

170. Refer XII NCERT Page No – 211, 212

172. Refer XII NCERT Page No – 73, 74

When marriage happens between colourblind woman and also suffering from sickle cell anaemia and non colourblind man and also suffering from sickle cell anaemia, all sons are colourblind with sickle cell anaemia.

174. Refer XII NCERT Page No – 182